THE ESSENTIAL IMAGE OF THE FUNCTOR OF POINTS COMPACT OPENS AND FINITE AFFINE ATLASES

1. The statement

We use all commutative rings with set carriers in $\mathcal{U}_\ell$ as affine tests. Their object type lies in $\mathcal{U}_{\ell+1}$ and their hom sets lie in $\mathcal{U}_\ell$. A functor takes values in sets in $\mathcal{U}_\ell$. All rings and ring maps are unital; the zero ring is allowed. For a ring A put $h_A(B) = \mathrm{Hom}(A, B)$.

We seek an intrinsic description of the essential image of $X \mapsto h_X$, where $h_X(B) = \mathrm{Hom}(\mathrm{Spec}\, B, X)$ and X is a small qcqs locally ringed lattice. Full faithfulness is the preceding comparison result. The new assertion is existence of a representing scheme. We follow the compact-open formulation of [2, 3], while constructing the resulting lattice directly from finite chart data.

Definition 1.1. A functor F is a *Zariski sheaf* if compatible elements on every finite basic cover of a test ring have a unique amalgam. Explicitly, if $\bigvee_i D(a_i) = \top$ in $\mathrm{Zar}(B)$, then

$$F(B) \longrightarrow \prod_i F(B[1/a_i]) \rightrightarrows \prod_{i,j} F(B[1/(a_i a_j)])$$

is an equaliser, with the canonical maps of localisations. Empty covers are included; in particular $F(0)$ is contractible.

Different presentations of the localisations are related by their universal properties. The sheaf condition can equivalently be stated for arbitrary choices of the corresponding localisation rings.

The Zariski lattices define a functor Zar on rings. For $h : A \to B$, its lattice map is characterised by

$$\mathrm{Zar}(h)(D(a)) = D(h(a)).$$

Uniqueness of this characterisation proves identity and composition.

Definition 1.2. A *compact open* of F is a natural transformation $U : F \to \mathrm{Zar}$. Its realisation is the subfunctor

$$|U|(B) = \sum_{x : F(B)} (U_B(x) = \top).$$

It is *affine* if there merely exist a small ring A and a natural equivalence $h_A \simeq |U|$. A finite family U_i covers F if $\bigvee_i U_i = \top$ pointwise. A *functorial qcqs scheme* is a Zariski sheaf merely admitting such a finite affine compact-open cover.

Compactness here is a condition on every affine pullback. Thus the compactness of the overlaps is built into the definition. We do not require the overlaps to be affine, nor add that as a gluing hypothesis.

Theorem 1.3 (Essential image)**.** *A small-set-valued functor F is a functorial qcqs scheme if and only if there merely exist a small qcqs locally ringed lattice X and a natural equivalence $F \simeq h_X$.*

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Together with full faithfulness, this gives the comparison on objects and arrows. To state an equivalence of univalent categories one must also package the geometric category and prove its univalence. That packaging is separate from Theorem 1.3.

2. Compact opens on affine tests

For any F, evaluation at 1_A gives the covariant Yoneda equivalence

$$\mathrm{Nat}(h_A, F) \simeq F(A).$$

Its inverse sends x to $h \mapsto F(h)(x)$. The two inverse laws are the identity law and the naturality equation at $h : A \to B$. In particular compact opens on h_A correspond to elements of $\mathrm{Zar}(A)$. For $u \in \mathrm{Zar}(A)$ the corresponding subfunctor is

$$B \longmapsto \{h : A \to B \mid \mathrm{Zar}(h)(u) = \top\}.$$

If $u = D(a)$, its realisation is $h_{A[1/a]}$, by the universal property of localisation and $D(h(a)) = \top$ if and only if $h(a)$ is a unit.

More generally, for $x \in F(A)$, pullback of $|U| \to F$ along the transformation $h_A \to F$ determined by x is the affine compact open classified by $U_A(x)$. This is a direct consequence of naturality:

$$U_B(F(h)(x)) = \mathrm{Zar}(h)(U_A(x)).$$

This formula is the precise link between the classifier definition and compact open subfunctors. No ideal representatives are chosen.

The compact opens of F form a distributive lattice with pointwise operations. This lattice initially lies in $\mathcal{U}_{\ell+1}$, because its elements are natural transformations on all small rings. Even though each value $\mathrm{Zar}(A)$ is small, the entire lattice must not simply be asserted small.

3. The geometric functor has an affine compact-open atlas

Let X be a small qcqs locally ringed lattice. For an open e of X define

$$U_e(p) = p^*(e), \qquad p : \operatorname{Spec} B \to X.$$

This is natural: the inverse-image map of $\operatorname{Spec} h$ is $\mathrm{Zar}(h)$, and inverse images compose. The slice universal property gives

$$|U_e|(B) \simeq \mathrm{Hom}(\operatorname{Spec} B, X|e).$$

Indeed a map factors uniquely through $X|e$ exactly when $p^*(e) = \top$; the factorisation preserves locality. Thus affine chart opens give affine compact opens, and a finite chart cover remains a cover.

The functor h_X is a Zariski sheaf: maps from the basic open pieces of $\operatorname{Spec} B$ glue uniquely by the finite-cover descent theorem for morphisms, using $(\operatorname{Spec} B)|D(a) \simeq \operatorname{Spec}(B[1/a])$. This includes the empty-cover case. Consequently h_X satisfies the left side of Theorem 1.3.

4. From a functorial atlas to gluing data

Assume temporarily that a finite affine atlas of F is given:

$$U_1, \ldots, U_n : F \to \mathrm{Zar}, \qquad h_{A_i} \simeq |U_i|.$$

Let $x_i \in F(A_i)$ be the resulting generic chart point, and put

$$d_{ij} = U_j(x_i) \in \mathrm{Zar}(A_i), \qquad X_i = \operatorname{Spec} A_i.$$

The preceding pullback calculation identifies $|U_i \wedge U_j|$ with the functor of points of $X_i|d_{ij}$. It also identifies it with that of $X_j|d_{ji}$. These open subspaces are qcqs, since compact opens of qcqs schemes have finite affine covers. Full faithfulness therefore produces transition isomorphisms

$$\theta_{ij} : X_i|d_{ij} \simeq X_j|d_{ji}.$$

First the classifiers of a third compact open show that θ_{ij} takes $X_i|(d_{ij} \wedge d_{ik})$ to $X_j|(d_{ji} \wedge d_{jk})$. On every affine probe the two inverse images of U_k agree, since the transition retains the same point of F. Affine probes detect equality of compact opens of qcqs spaces: on an affine atlas one tests with its inverse canonical affine isomorphisms, then uses lattice separation on the atlas. This proves the domain equation.

Now restrict the transitions to those identified triple domains. The two composites into the third chart give the same point of F on every affine probe. A chart inclusion into F is an embedding, so the composites agree as chart points; faithfulness makes them equal as geometric maps. This is the cocycle law on its specified domain. Identity and inverse laws follow by the same argument. No choice of generators for the overlap opens is involved.

The restricted cocycle and its section equation. Here are the domains and restrictions in the preceding argument explicitly. Use the contravariant indices of the gluing construction: put $Y_{ij} = X_j|d_{ji}$, let $h_{ij} : Y_{ij} \to X_i$ be the transition followed by inclusion, and write $t_{ij}(a)$ for its inverse image, viewed in L_j. Its image lies below d_{ji}. The transition square says that h_{ij} and the inclusion of Y_{ij} into X_j induce the same point of F on every affine probe.

For three indices set $q = t_{jk}(d_{ji})$. The already proved domain equation, after including the slice lattice into L_k, is

$$q = d_{kj} \wedge d_{ki}.$$

In particular $q \leq d_{kj}, d_{ki}$. Restrict h_{jk} to $X_k|q$; its inverse image of d_{ji} is top, so it factors uniquely through Y_{ij}. Call this factor a. There are now two actual maps with the same source and target,

$$h_{ij}a, \quad h_{ik}\big(X_k|q \longrightarrow Y_{ik}\big) : X_k|q \longrightarrow X_i.$$

Their points of F agree by applying the transition square twice on the left and once on the right; the factorisation triangles and associativity identify the intermediate chart point. The final chart inclusion is injective on affine points. Since $X_k|q$ is qcqs, affine probe separation makes the displayed maps equal.

Taking inverse images of $b \in L_i$ yields

$$t_{jk}(t_{ij}(b)) \wedge q = t_{ik}(b) \wedge q.$$

The left meet simplifies by monotonicity from $t_{ij}(b) \leq d_{ji}$. On the right use $q = d_{kj} \wedge d_{ki}$ and $t_{ik}(b) \leq d_{ki}$. This proves the exact lattice cocycle $t_{jk}(t_{ij}(b)) = d_{kj} \wedge t_{ik}(b)$.

For sections, apply equality of the two maps after restricting their images to the common open $w = t_{jk}(t_{ij}(b)) \leq q$. On the left, naturality of h_{jk} and composition of restrictions remove the restriction introduced by the factorisation. On the right, composition of restrictions gives the restriction of $c_{ik,b}(s)$ directly to w. Thus

$$c_{jk,t_{ij}(b)}(c_{ij,b}(s)) = c_{ik,b}(s)|_w.$$

In the overlap slice Y_{jk}, the restriction on the right is the restriction first to $t_{ik}(b) \wedge d_{kj}$ and then along its equality with w. Composition of restrictions identifies these expressions. This is precisely compatibility in the equaliser defining the glued section ring; no equality of section carriers is presumed.

For a repeated chart, $d_{ii} = \top$: its generic point belongs to U_i. The transition square and chart injectivity give h_{ii} equal to the open inclusion. Inverse images and restriction of section

images give $t_{ii}(b) = b$ and the section identity. Applying the same point argument to the reverse transition gives its composite with the forward transition equal to identity; cancellation of the final open inclusion gives the inverse equation in the overlap space itself.
Review before formalisation. The source $X_k|q$ was fixed before comparing either composite. Factoring through Y_{ij} uses $q = t_{jk}(d_{ji})$, whereas restricting the direct transition uses $q \leq d_{ki}$; these are distinct uses of the domain equation. Section equality is always stated after restriction to a common open, so no transport of ring carriers is implicit. The triple source is qcqs by restriction of the affine X_k, even when it is not affine. Finally an actual atlas still includes only merely given affine witnesses: finite choice at universe $\ell + 1$ supplies all these witnesses merely, and both this truncation and the atlas truncation are eliminated only into mere representability. Empty finite families are allowed throughout.

5. A small finite gluing construction

Here is the finite gluing theorem needed in the reconstruction, in the ringed-lattice approach of [1]. Let X_i have small lattices L_i, let $d_{ii} = \top$, and suppose the transition isomorphisms above are given with the domain and cocycle laws. Define

$$L = \left\{ (v_i) \in \prod_i L_i \;\middle|\; \theta^*_{ij}(v_j \wedge d_{ji}) = v_i \wedge d_{ij} \text{ for all } i, j \right\}.$$

Both sides are viewed in L_i. Operations are componentwise. The transition maps preserve the operations on the restricted lattices, so the displayed equations are stable under joins, meets, bottom and top. The equations are propositions and L is a small set.

Here is a precise form of the transition laws used in this construction. Restriction followed by transport defines a bounded lattice map

$$r_{ij} : L_i \longrightarrow L_j|d_{ji}.$$

Write t_{ij} for its underlying function into L_j. The identity and domain-sensitive cocycle equations are

$$t_{ii}(a) = a, \qquad t_{jk}(t_{ij}(a)) = d_{kj} \wedge t_{ik}(a).$$

In particular $t_{ji}(t_{ij}(a)) = d_{ij} \wedge a$. These equations include the domains of the compositions; an equation of transitions on an unspecified triple intersection would not suffice. The corresponding local ring maps are

$$c_{ij,a} : \mathcal{O}_i(a) \longrightarrow \mathcal{O}_j(t_{ij}(a)).$$

Their identity equation says that $c_{ii,a}(s)$ restricts back to s along $t_{ii}(a) = a$. Their cocycle equation says

$$c_{jk,t_{ij}(a)}(c_{ij,a}(s)) = c_{ik,a}(s)|_{\,t_{jk}(t_{ij}(a))},$$

where the restriction is defined by the preceding domain equation. Thus the transported sections form a compatible tuple. These lattice and section equations are sufficient for the chart argument below.

For $v = (v_i)$, define $\mathcal{O}(v)$ to be the ring of tuples $s_i \in \mathcal{O}_i(v_i)$ whose restrictions agree under the transition maps on $v_i \wedge d_{ij}$. This is an equaliser of maps of products of small rings. Restriction and ring operations are componentwise. Sheaf gluing is componentwise; agreement of the glued components follows by separation on the overlap. Invertibility opens are

$$D_v(s) = (D_{v_i}(s_i))_i.$$

Locality of the transitions makes this an element of L. A compatible tuple is a unit precisely when its components are units: their inverses are compatible by uniqueness of inverses. This proves the defining property of the invertibility open, and the support laws hold componentwise.

The ith chart open has jth component d_{ji}. Its membership in L uses the domain equations of the transitions, and the charts cover because $d_{jj} = \top$. Projection identifies its lower interval with L_i. The inverse takes $a \in L_i$ to the family whose jth component is the image of $a \wedge d_{ij}$ under the inverse transition on opens. The domain and cocycle equations prove compatibility of that family. They also show that a section tuple on this chart is uniquely determined by its ith component. We obtain $X|e_i \simeq X_i$, with the prescribed transition maps. Thus X is a small qcqs scheme.

This construction uses only finite products and equaliser conditions in $\mathcal{U}_\ell$. In particular it does not construct a large locally ringed lattice and then silently resize it. For an empty atlas the empty products give the one-element lattice and zero section ring, hence the empty scheme; its value on the zero test ring is a singleton.

6. Identifying the represented functor

We need the following descent principle in the category of Zariski sheaves. If V_i is a finite compact-open cover of a functor G, and H is a Zariski sheaf, then compatible natural maps $|V_i| \to H$ glue uniquely to a map $G \to H$.

To define its value at $x \in G(B)$, choose finite basic presentations of the compact opens $V_i(x)$, whose union covers $\operatorname{Spec} B$. On each basic piece, x becomes an element of the relevant $|V_i|$. Apply the given local map, and use compatibility on intersections to glue in H. Uniqueness is sheaf separation. The type of such an amalgam is a proposition before a merely existing presentation is eliminated. Naturality follows by restricting this characterisation along a ring map and testing on the pulled-back basic cover. This proves the descent principle without global choices.

More explicitly, a value $y \in H(B)$ fits $x \in G(B)$ when, for every $h : B \to C$ and every i with $G(h)(x) \in |V_i|(C)$,

$$H(h)(y) = \alpha_i(G(h)(x)).$$

Two fitting values agree on any finite basic refinement of the $V_i(x)$, hence agree by separation. Existence is first proved using an actual refinement and then descended to this proposition of fitting candidates. The base-change square

$$\begin{array}{ccc} B & \longrightarrow & B[1/a] \\ \downarrow & & \downarrow \\ C & \longrightarrow & C[1/h(a)] \end{array}$$

comes directly from the universal property of localisation. It compares the local amalgam equations after changing the test ring. This proves fitting for all h, and fitting in turn proves naturality by uniqueness. Consequently the type of natural extensions is contractible.

Equality of maps into a sheaf can therefore be tested on a finite compact-open cover. In particular compatible local natural equivalences, whose inverses are also compatible on overlaps, glue to a natural equivalence: the two composites are identities on each member of the respective covers, hence identities globally.

The charts of the constructed X and those of F now give maps between F and h_X: apply the descent principle in each direction. The overlap conditions are exactly the transition equations already established. The two composites restrict to identities on all charts, hence are identities by uniqueness. Therefore $F \simeq h_X$.

Finally, the atlas of F merely exists. An actual atlas gives a witness of representation; insert it into

$$\left\| \sum_X \big(X \text{ is small and qcqs}\big) \times (F \simeq h_X) \right\|.$$

This is a proposition, so the mere atlas can be eliminated into it. We do not claim an untruncated canonical geometric object without the additional univalence theorem for the geometric category.

7. Over a base ring

Fix a small commutative ring k with set carrier. Relative functors are native functors on actual k-algebra tests $(B, \iota : k \to B)$ and algebra maps. Their sheaf condition uses finite basic algebra diagrams; their compact opens are natural maps to the restriction of Zar, and their affine charts are represented by small k-algebras. These are intrinsic relative conditions, not definitions in terms of an absolute total functor.

Define

$$(\mathsf{T}F)(B) = \sum_{\iota : \mathrm{Hom}(k,B)} F(B, \iota).$$

For $h : B \to C$ let $c_{h,\iota} : (B, \iota) \to (C, h\iota)$ be its canonical algebra lift. The total action sends (ι, x) to $(h\iota, F(c_{h,\iota})(x))$. The dependent comparison needed for every subsequent argument is

$$\mathrm{transport}_{j \mapsto F(C,j)}(p, F(c_{h,\iota})(x)) = F(h, p)(x), \qquad p : h\iota = \kappa.$$

Path induction proves it: at reflexivity the algebra maps have the same underlying ring map, and their commuting proofs are propositions. Consequently total reindexing along any actual algebra map a satisfies

$$(\mathsf{T}F)(\underline{a})(\iota, x) = (\kappa, F(a)(x)).$$

The total identity and composition laws follow from this whole-pair equation and the native functor laws. Projection gives a natural map $\pi_F : \mathsf{T}F \to h_k$.

Theorem 7.1 (Intrinsic relative totalisation)**.** *A native relative functor F is a Zariski sheaf if and only if $\mathsf{T}F$ is an absolute Zariski sheaf. It is functorially qcqs if and only if $\mathsf{T}F$ is absolutely functorially qcqs.*

Proof. For a compatible total cover family (ι_i, x_i), first glue the base maps in the representable sheaf h_k. Write the resulting equations as $p_i : \lambda_i\iota = \iota_i$. Transport x_i backwards along p_i into $F(B_i, \lambda_i\iota)$. Their pairs equal the original total points. The whole-pair reindexing equation above transfers compatibility; injectivity of pairing at the fixed overlap base gives native relative compatibility. Glue the rebased values in F. For uniqueness, any other total amalgam has the same base map by uniqueness in h_k; rebase its value along that equation and apply relative uniqueness. Conversely, total gluing of paired native values and separation in h_k rebase the glued point into the specified fibre. Total uniqueness and fixed-base injectivity give relative uniqueness. Empty covers are included.

Compact opens and their cover equations pass between the relative and total functors by evaluation at (ι, x). Reordering the membership proof gives $\mathsf{T}|U| \simeq |\mathsf{T}U|$. For an algebra (A, α), forgetting the structure map gives $\mathsf{T}h^k_{(A,\alpha)} \simeq h_A$: that structure is forced to be $h\alpha$ by its underlying ring map $h : A \to B$. Thus relative affine atlases totalise. Conversely a chart $e : h_A \simeq \mathsf{T}G$ supplies $\alpha : k \to A$ from the base component of $e_A(1_A)$. Naturality says that the base of $e_B(h)$ is $h\alpha$. Transporting along that equation restricts e to a native relative affine equivalence. Applying this to the total chart realisations recovers a relative affine atlas. Atlas and affine-witness truncations are eliminated only into propositions. □

Theorem 7.2 (Relative representation and categorical comparison)**.** *A native functor F on small k-algebras is functorially qcqs if and only if there merely exist a small geometric qcqs space X, a map $s : X \to \operatorname{Spec} k$, and a native natural equivalence*

$$F(B, \iota) \simeq \operatorname{Hom}_{\operatorname{Spec} k}(\operatorname{Spec}(B, \iota), (X, s)).$$

The relative functor of points is an equivalence of univalent categories between geometric qcqs schemes over $\operatorname{Spec} k$ *and intrinsic relative functorial qcqs schemes. It has an explicit inverse functor and both natural isomorphisms.*

Proof. Totalise F and apply the absolute representation theorem. For an actual representation $e : \mathsf{T}F \simeq h_X$, compose e^{-1} with π_F to get $q : h_X \to h_k$. The affine comparison and absolute full faithfulness give $s : X \to \operatorname{Spec} k$ and the equation $sp = \operatorname{Spec}(q_B(p))$ on every affine probe. Taking fibres over ι therefore gives geometric points over k. On the original side the fibre is

$$\sum_{\kappa : k \to B} \sum_{x : F(B,\kappa)} (\kappa = \iota) \simeq F(B, \iota),$$

where the equivalence transports x along the displayed path. Path induction gives its two inverse laws. Restricting e to fibres and composing these equivalences gives the native relative representation. The mere absolute representation is eliminated only into mere relative representability. Conversely, the total relative geometric functor is h_X, by the canonical lift and uniqueness of the base map of an ordinary affine point. The converse totalisation criterion and invariance under relative natural equivalence prove the necessary direction.

The geometric category over k is a univalent slice. Native relative functors form a univalent category by the value-family and action identity argument; the relative qcqs condition is a proposition. Existing relative full faithfulness and the just-proved relative essential image give the categorical comparison by the same propositional-fibre argument as in the next section. □

The companion note *Relative totalisation over a base ring: a proof with explicit changes of structure map* gives the detailed argument, its review, and the declaration-by-declaration interfaces used to handle the dependent types. It was written before formalising this continuation.

8. Equivalence of univalent categories

This section gives the categorical conclusion, including its univalence arguments. The structured identity principles, category packaging, comparison functor and explicit quasi-inverse are now checked, as well as the preceding absolute Hom–Nat and essential-image theorems. The informal proof and its review were written before formalisation.

Let $\mathcal{C}$ have objects small locally ringed lattices X together with a proof that X is qcqs, and local geometric maps as arrows. Let $\mathcal{D}$ have objects small-set-valued functors F together with a proof that F is functorially qcqs, and natural transformations as arrows. Their identities and compositions are the existing geometric and natural ones. Geometric maps and natural transformations form sets. Both object types lie in $\mathcal{U}_{\ell+1}$; geometric homs lie in $\mathcal{U}_\ell$, whereas natural transformations are initially in $\mathcal{U}_{\ell+1}$. All categories can consequently be considered at the common bound $\ell + 1$. No resizing of the natural-transformation types is needed.

The functor of points is

$$H : \mathcal{C} \longrightarrow \mathcal{D}, \qquad H(X) = h_X, \quad H(f)_B(p) = f \circ p.$$

The geometric atlas theorem supplies its object property. The unit and associativity equations prove that it is a functor. The checked Hom–Nat theorem says that H is fully faithful, and the checked essential-image theorem says that it is merely essentially surjective.

Lemma 8.1 (Univalence of the geometric category)**.** *For $X, Y \in \mathcal{C}$, the canonical map*

$$(X = Y) \longrightarrow \mathrm{Iso}_{\mathcal{C}}(X, Y)$$

is an equivalence.

Proof. First consider commutative rings with set carriers and bounded distributive lattices. Univalence identifies equality of their carriers with equivalence of carriers. Under this identification, transport of operations is exactly preservation of those operations. Function extensionality identifies the corresponding equations of operations; the equational axioms and sethood proofs are propositions. This proves the structured identity principle for these two algebraic structures: paths correspond to algebra isomorphisms, with the correspondence being the canonical one at reflexivity.

For a fixed lattice L, consider presheaves of rings with set carriers. An identity of their ring families corresponds to a family of ring isomorphisms. After transporting along that identity, equality of the restriction maps is exactly naturality of the family of isomorphisms. Restriction laws and the sheaf condition are propositions. Thus equality of sheaves of rings on L corresponds to natural ring isomorphisms. This assertion follows by equivalence induction on the ring families: in the identity case naturality states equality of restrictions, and function extensionality gives the equality of restriction families.

Now an isomorphism $f : X \to Y$ gives a lattice isomorphism $f^* : L_Y \to L_X$ and ring isomorphisms

$$f_u^\sharp : \mathcal{O}_Y(u) \longrightarrow \mathcal{O}_X(f^* u)$$

compatible with restrictions. Conversely, such data give an isomorphism of ringed lattices. Any such isomorphism between locally ringed lattices is local: ring isomorphisms preserve and reflect units, so the defining universal property of invertibility opens identifies the two opens. Its inverse is local by the same argument.

Use the inverse lattice isomorphism to identify L_X with L_Y. With this base fixed, the preceding sheaf identity principle identifies the sheaves using $f^\sharp$ (in the appropriate inverse orientation). For a fixed sheaf, the type of invertibility maps is a proposition by uniqueness of invertibility opens; its support laws are propositions too. Consequently the locally ringed structures are identified, including these fields. Finally the qcqs witnesses lie in a proposition, so they add no further identity data.

These identifications are equivalences at each stage of the dependent record, rather than just a construction of some path from an isomorphism. They agree with the identity isomorphism at reflexivity. Path induction therefore identifies their composite with the canonical map from paths to geometric isomorphisms. This proves the required univalence. □

Lemma 8.2 (Univalence of the functorial category)**.** *For $F, G \in \mathcal{D}$, the canonical map*

$$(F = G) \longrightarrow \mathrm{Iso}_{\mathcal{D}}(F, G)$$

is an equivalence.

Proof. A categorical isomorphism of set-valued functors is exactly a natural transformation with equivalence components: its pointwise inverse is natural by injectivity of the forward components. Apply univalence pointwise to identify the value sets along those component equivalences, and function extensionality to identify the value families. After this identification, naturality is exactly equality of the reindexing maps. The sethood proofs and functor laws are propositions. This gives the structured identity principle for set-valued functors, with canonical map the identity-to-natural-isomorphism map. The functorial qcqs condition is a proposition, so restriction to that full subcategory preserves univalence. □

Theorem 8.3 (Categorical comparison)**.** *The functor $H : \mathcal{C} \to \mathcal{D}$ is an equivalence of univalent categories. Its map on object types is an equivalence.*

Proof. Full faithfulness induces equivalences on isomorphism types. Indeed, lift an isomorphism and its inverse by fullness; their inverse equations hold by faithfulness. These lifts are unique by the equivalences on hom sets. For $X, Y \in \mathcal{C}$ we therefore have the commutative square

$$\begin{array}{ccc} (X = Y) & \xrightarrow{\operatorname{ap} H} & (HX = HY) \\ \downarrow\simeq & & \downarrow\simeq \\ \operatorname{Iso}_{\mathcal{C}}(X, Y) & \xrightarrow{\simeq} & \operatorname{Iso}_{\mathcal{D}}(HX, HY). \end{array}$$

Commutativity is path induction; the vertical maps are univalence and the bottom map is the preceding equivalence on isomorphisms. Thus $\operatorname{ap} H$ is an equivalence: the object map of H is an embedding.

For $F \in \mathcal{D}$, its fibre

$$P_F = \sum_{X:\mathcal{C}} (HX = F)$$

is consequently a proposition. The essential-image theorem gives merely an X and a natural equivalence $F \simeq HX$. Invert that equivalence and apply univalence of $\mathcal{D}$ to get $\|P_F\|$. Since P_F is a proposition, eliminate this truncation into P_F itself. It is therefore inhabited and contractible. Every fibre of the object map is contractible, so H is an equivalence on objects. This is the precise step that removes the truncation; it uses univalence and full faithfulness, not a choice of affine atlases.

For completeness, its categorical inverse can be read off from these fibres. Write their centres as $(KF, \epsilon_F : HKF = F)$ and let $e_F : HKF \cong F$ be the resulting isomorphism. For $\alpha : F \to G$, define $K\alpha$ as the unique geometric arrow with

$$H(K\alpha) = e_G^{-1} \circ \alpha \circ e_F.$$

Existence and uniqueness use the hom equivalence. Faithfulness proves the identity and composition laws. The e_F form a natural isomorphism $HK \cong 1_{\mathcal{D}}$ by this defining equation. Lifting e_{HX} gives a natural isomorphism $KH \cong 1_{\mathcal{C}}$; it is an isomorphism since full faithfulness reflects isomorphisms. Hence K is a quasi-inverse functor, as claimed. □

Review. The geometric univalence proof includes the ring-family and restriction maps, not just the lattice. The inverse orientation of inverse images is respected. Invertibility maps are unique by their universal property; they must not be treated as arbitrary extra operations. Adding a merely existing affine atlas as a property introduces no chosen cover. Both univalence statements assert equivalence of identity types with isomorphisms, rather than only that isomorphic objects can be identified. In particular automorphisms remain as loops of the object types. The representation fibres are propositions only after full faithfulness and univalence have been established. The resulting removal of truncation uses no axiom of choice and entails no universe resizing.

9. Review and connection with the implementation

The proof separates four interfaces: functors on all small tests and Yoneda; compact opens and their realisations; finite compact-cover descent for maps into sheaves; and finite geometric gluing with cocycles. Compact-open atlases account for quasi-separatedness. The finite tuple construction accounts for smallness. Mere existence of the atlas is used only after truncating the representation statement. None of these points is supplied by full faithfulness alone.

The existing checked development supplies full faithfulness, geometric finite-cover descent, restriction to compact opens, and their qcqs property. The first new checkpoint now supplies the arbitrary-functor and compact-open interfaces described above. In particular, `TestFunctors`, `TestYoneda`, and `TestEquivalences` give the functor interface on all small rings; `ZariskiFunctor` gives the classifier and its comparison with inverse images under Spec. `CompactOpens`, `OpenRealisation`, and `CompactPullback` give the lattice, realisation and affine pullback formula. `GeometricOpens` proves the realisation equation for geometric opens. `TestSheaves` states the finite basic sheaf condition, including empty covers. `RepresentableSheaf` proves it for all affine representables using the earlier localisation gluing theorem. `ZariskiSeparated` proves separation of the classifier; `OpenSheaf` proves that compact-open realisations of sheaves are sheaves. `WholeRealisation` and `SheafTransfer` supply the whole-open equivalence and invariance under natural equivalence.
`FunctorialQcqs` gives the intrinsic definition. `BasicAffineFunctor` identifies $|D(a)|$ with $h_{A[1/a]}$; `CompactAffineCover` proves that every compact open of an affine representable is a functorial qcqs scheme.
`GeometricEquivalence` and `AffineFunctor` transfer geometric isomorphisms and affine objects to functors. `SpaceGluing` states geometric finite-cover descent directly in terms of maps agreeing on intersections. `OpenMonic` and `BasicGeometry` identify the geometric basic-open restriction diagram with the localisation diagram. `GeometricSheaf` then proves sheafhood of h_X for every locally ringed lattice X. Together with the atlas construction, `GeometricAtlas.qcqs` proves that every geometric qcqs scheme gives a functorial qcqs scheme.
`FiniteLatticeGlue` and `FiniteRingGlue` construct the small compatible-tuple lattice and rings with their universal properties; units are detected coordinatewise. `FiniteRingedGlue` supplies the restriction maps and sheaf proof, and `FiniteLocalGlue` supplies the locally ringed structure, local projection maps, and their equality on overlaps.
`FiniteTransitionLattice` proves the covering equation and the lattice chart theorem under the displayed domain-sensitive laws. `FiniteTransitionGlue` proves the section-ring chart theorem, the geometric chart isomorphisms, and the qcqs conclusion for affine input charts. The extraction of these hypotheses from a functorial atlas is now checked in the atlas modules described below.
`LocalisationTests` and `CompactMembership` give the base-change square and local membership criterion used in descent. `CompactCoverDescent` proves contractibility of natural extensions, with no sheaf assumption on the source. `CompactCoverSeparation` gives local detection of equality, and `LocalFunctorIso` glues compatible local natural equivalences of sheaves.
The overlap construction is also checked. `RealisationTransfer` restricts a natural equivalence using a membership equivalence; `OpenIntersections` and `OpenSymmetry` identify realised intersections while retaining their underlying point. `AffineOpenRealisation` represents every compact open of h_A by the corresponding open subspace of $\operatorname{Spec} A$. `AtlasOverlap` therefore represents affine-chart overlaps by qcqs spaces and records their inclusion into the original functor. `GeometricRecovery` recovers geometric isomorphisms from natural equivalences, and `AtlasTransition` gives the transition isomorphisms and their equations over that functor. `GeometricOpenSeparation` shows that affine probes detect equality of compact opens of qcqs spaces. Using it, `AtlasDomains` proves that a transition identifies the two inverse images of any third compact open, hence the triple-overlap domain equation.

One change of presentation is treated explicitly. `SpacePaths` transports a diagram along equalities of its spaces and proves that composition and isomorphisms are preserved. The basic-open diagram is checked at its defining presentation and moved as a whole. This avoids rechecking the component proofs of the lattice and sheaf when packaging the diagram in geometric notation.

The absolute essential-image theorem is now checked. `AtlasInverse` proves the diagonal and reverse-transition laws. `OpenFactor` restricts and factors maps through an open with an explicit lattice and section interface. `AtlasData` packages the actual finite chart data and the domain equations. `AtlasCocycle` compares the two actual maps on $X_k|q$; `AtlasLaws` and `AtlasSections` extract exactly the lattice and section hypotheses of finite gluing. `AtlasReconstruction` applies that theorem and records the new chart classifiers. `AtlasCompatibility` proves compatibility in both directions, and `AtlasRealisation` applies the local-to-global equivalence result.

For clarity, the compatibility argument has two directions. If points p in chart i and r in chart j have the same F-point, the classifier puts r in d_{ji}. Factor r through that overlap. Its transition image equals p by injectivity of the original chart. The equality of the two overlap maps into the glued space then gives equal geometric points. Conversely, if their geometric points agree, the reconstructed classifier puts r in d_{ji}. The same overlap square and injectivity of the reconstructed chart identify its transition image with p; the transition trace gives equal F-points. Thus neither inverse compatibility nor agreement of classifiers is assumed in applying the local-to-global equivalence.

`FiniteLargeChoice` proves finite choice for large affine witnesses indexed by the existing small finite type; finite types at different universe levels are not silently identified. `FunctorialTransfer` transfers sheafhood and affine atlases across a natural equivalence. `EssentialImage.reconstruct` eliminates the finite witnesses and the atlas into mere representability; `recognise` proves the converse. `equivalence` identifies the two propositions in Theorem 1.3. The relative totalisation and representation theorems of Section 7 are now checked too, including the equivalence of relative univalent categories. The absolute categorical comparison of Section 8 remains the input to that reduction.

The informal restricted-cocycle proof and review were written before formalisation. During the opening audit the section-composite argument was separated into its stated intermediate section images, and the factor map was made opaque after its lattice and restriction equations. The reconstruction triangle is checked at its defining glued space. These changes keep the formal proof aligned with its mathematical steps. The categorical continuation reuses `SipGeneric` and the existing `Category`, `Arrows`, `Functor`, and `EquivalenceLemmas` interfaces without changing them. `RingIdentity` and `LatticeIdentity` apply the generic structure identity theorem to the actual algebra records. `FamilyIdentity` and `DecoratedIdentity` pass identity systems to families, structure, and propositional properties. `PresheafIdentity` and `LocalIdentity` apply these results to rings with restriction maps and their locally ringed properties. `SpaceIdentity` handles the dependent lattice/structure pair; `SpaceIsoIdentity` identifies the structural presentation of an actual ringed-lattice isomorphism with its original ringed map. `GeometricCategory` uses this bridge and the categorical singleton criterion to prove univalence for all spaces.

`SetIdentity` and `FunctorIdentity` prove the analogous identity principle for the existing set-valued functors; `FunctorCategory` proves categorical univalence. `FullSubcategory` restricts both categories to their qcqs properties. `WeakCategoryEquivalence` proves the object embedding and removes the truncation of essential surjectivity using

`EmbeddingFibres`. `CategoryQuasiInverse` constructs the inverse functor and its two natural isomorphisms, with checked identity, composition, and naturality laws. `QcqsCategories.equivalence` assembles the categorical equivalence for the existing functor of points. Its source and target are actual univalent category records; their univalence proofs are constructed in this continuation, rather than parameters assumed by the comparison. `object-equivalence` separately records the equivalence on object types. These conclusions use the existing universe bounds and no axiom of choice or resizing.
The relative continuation follows the detailed companion note.
`RelativeTotal.change-target` states the dependent transport comparison, and `on-algebra` compares the whole total point for an arbitrary algebra map.
`RelativeCoverFibre` separates pair equalities from native fibre equalities; `RelativeBaseGlue`, `RelativeTotalSheaf`, and `RelativeSheafOfTotal` implement the two sheaf arguments. `RelativeTotalisation` exposes both the sheaf and qcqs equivalences, with intrinsic native definitions. `RelativeEssentialImage` proves relative reconstruction and its converse. `RelativeCategories.Over[K].equivalence` packages the relative categorical comparison, using `SliceCategory` and `RelativeFunctorCategory` for univalence.
The complete qcqs tree now checks with 249 modules and 9,250 declarations, zero failures. All 3,752 lists in the combined absolute, category, and relative opening audit report no extra openings. No construction proof of function extensionality, localisation, or quotients is reopened by the relative clients. The detailed note names the intermediate base paths, rebased values, and pointwise squares that keep the checker aligned with the mathematical argument.
The approach is the compact-open formulation of [2, Sections 4–5] and [3, Sections 5–6]; its finite small gluing construction replaces the universe-lifted realisation route.

References

[1] Thierry Coquand, Henri Lombardi and Peter Schuster, *Spectral schemes as ringed lattices*, Annals of Mathematics and Artificial Intelligence **56** (2009), 339–360.
[2] Max Zeuner, *Univalent Foundations of Constructive Algebraic Geometry*, Sections 4–5. Local source: `SCHEME/zeuner1.pdf`.
[3] Max Zeuner and Matthias Hutzler, *The Functor of Points Approach to Schemes in Cubical Agda*, Sections 5–6. Local source: `SCHEME/zeuner3.pdf`.